

Counting $Z\varphi$ -classes for PerfectGroup(7680, 4)

Claude Archer

n	$ S $	$ E $	#SG($8n$)	# $Z\varphi$	cumulative	time (s)	cum. time (s)
2	16	15 360	14	16	16	1.215	1.215
3	24	23 040	15	1	17	0.476	1.691
4	32	30 720	51	224	241	8.117	9.808
5	40	38 400	14	1	242	0.560	10.368
6	48	46 080	52	32	274	5.301	15.669
7	56	53 760	13	1	275	0.518	16.187
8	64	61 440	267	3 249	3 524	88.368	104.555
9	72	69 120	50	2	3 526	3.119	107.674
10	80	76 800	52	32	3 558	4.882	112.556
11	88	84 480	12	1	3 559	0.483	113.039
12	96	92 160	231	792	4 351	50.028	163.067
13	104	99 840	14	1	4 352	0.551	163.618
14	112	107 520	43	32	4 384	4.386	168.004
15	120	115 200	47	1	4 385	2.148	170.152
16	128	122 880	2328	72 428	76 813	2240.689	2410.841
17	136	130 560	15	1	76 814	3.152	2413.993
18	144	138 240	197	80	76 894	24.052	2438.045
19	152	145 920	12	1	76 895	0.422	2438.467
20	160	153 600	238	808	77 703	61.154	2499.621
21	168	161 280	57	2	77 705	3.808	2503.429
22	176	168 960	42	32	77 737	4.871	2508.300
23	184	176 640	12	1	77 738	0.592	2508.892
24	192	184 320	1543	18 986	96 724	903.560	3412.452
25	200	192 000	52	2	96 726	3.879	3416.331
26	208	199 680	51	32	96 758	5.862	3422.193
27	216	207 360	177	5	96 763	15.086	3437.279
28	224	215 040	197	788	97 551	48.478	3485.757
29	232	222 720	14	1	97 552	0.615	3486.372
30	240	230 400	208	64	97 616	23.323	3509.695
31	248	238 080	12	1	97 617	0.559	3510.254
Total				97 617	97 617	3510.254	3510.254

Values of n contributing the largest shares of the total runtime

n	time (s)	share of total time	# $Z\varphi$	share of all $Z\varphi$ -classes
16	2240.689	63.83%	72 428	74.20%
24	903.560	25.74%	18 986	19.45%
8	88.368	2.52%	3 249	3.33%
20	61.154	1.74%	808	0.83%
Top four combined		93.83%		97.80%

Audited memory-optimized count-only computation, $2 \leq n \leq 31$

Structural and computational remarks. $\text{Out}(P) \simeq V_4$ does not act faithfully on $Z(P) \simeq C_4 \times C_2$: the image is C_2 and the kernel is C_2 . The auxiliary group S has order $8n$, and the resulting extensions have order $7680n$. All values in the first table were obtained with the same audited **count-only** implementation. The implementation is optimized primarily for memory: after a SmallGroup and the relevant subgroups/orbits have been processed, the constructed group objects are discarded rather than retained in a global list; only compact counting data are accumulated.

The second table shows that the runtime is strongly concentrated on the values of n that also contribute most of the $Z\varphi$ -classes. For the four largest runtime contributors together, the share is 93.83% of the total measured runtime and 97.80% of all $Z\varphi$ -classes for $2 \leq n \leq 31$.